Kiera Nechala

Mechanical Engineering
Varsity Swimmer

knechala@uwaterloo.ca linkedin.com/in/kiera-nechala-740672426

Summary of Qualifications

Brings hands-on engineering experience from a co-op at Clauger Canada, developing technical documentation and coordinating across teams, along with collaborative and technical skills built through the Autonomous Domino Placing Robot project (proficient in AutoCAD, SolidWorks, and MATLAB). Competing on the Varsity Swim Team reflects strong discipline, time management, and work ethic, qualities that translate directly into reliable, team-oriented performance on the job.

Education

Candidate for Bachelors of Mechanical Engineering **Expected 2030**
University of Waterloo *Scholarship: President's Scholarship of Distinction*
Courses: Advanced Calculus, Statistics for Engineers, Electromechanics, Materials Science

Experience

Engineering & Project Manager Intern, Clauger Canada Inc. **May – Sept 2026**

- Served as **Lead Project Manager** on a **$150,000** industrial refrigeration compressor replacement, coordinating scope, scheduling, and client communication.
- Authored a formal **TSSA variance application** justifying **CSA B51** clearance non-compliance for a Lochinvar pool heater installation (Oakridge Pool Boiler Replacement, Project No. 7749).
- Produced and submitted **9 AutoCAD drawings** supporting construction schedules and equipment specifications across multiple concurrent equipment-replacement projects.
- Interpreted **P&ID** drawings and TSSA piping schematics to assess ammonia refrigeration system compliance and equipment compatibility.

Projects

Ammonia Refrigeration System Redesign **Aug – Sept 2026**
Clauger Canada Inc. *London, ON*

- Conducted on-site field survey of a **100 TR ammonia refrigeration system**, capturing measurements and photos to establish as-built conditions for redesign.
- Designed mechanical layout and **P&ID in AutoCAD**; manually calculated refrigeration tonnage load to identify three compressors meeting system requirements.
- Identified an oversized compressor base against standard spec and authored a **variance report** to reduce footprint and cost, reviewed by senior engineering staff.

Autonomous Domino Placing Robot **Jan – Apr 2026**
University of Waterloo *Waterloo, ON*

- Programmed five autonomous placement algorithms in **C++** for a **VEX IQ** robot, enabling dynamic path-planning for **1–20** dominoes per run.
- Directed **engineering drawings**, tolerance testing, and physical prototype validation, achieving a **19–20/20** domino placement success rate at **under 5 seconds** per domino.
- Led iterative testing across mechanical, electrical, and software subsystems, contributing to the design meeting all **8 engineering specifications**, including a **92%** user-interface rating and **$24.05** fabrication cost (under a **$40** budget).

Technical Skills

- **Programming**: Python, C++, MATLAB, HTML/CSS
- **CAD & Engineering Software**: SolidWorks, AutoCAD, Fusion 360, Revit
- **Manufacturing & Prototyping**: Lathe, Milling Machine, Drill Press, Bandsaw
- **Engineering Fundamentals**: Engineering Drawing, GD&T Basics, Measurement & Tolerances

Extracurriculars

Varsity Swimming, University of Waterloo Varsity Sports Team **Sept 2025 – Present**
Training 20+ hours per week *Award: Swim Team, Female Rookie of the Year*

Certifications

- **Private Pilot License (PPL)**, Transport Canada *(In Progress)*
- **CSWA**, Certified SOLIDWORKS Associate
- **DELF B2**, Independent User of French